$$h[n] = \delta[n] - \delta[n-1]$$

ve

$$\delta[n] \longleftrightarrow 1$$

$$\delta[n-1] \longleftrightarrow e^{-j\omega}$$

$$\Rightarrow H(e^{j\omega}) = 1 - e^{-j\omega}$$

$$= \frac{2\,(1 - e^{-j\omega})}{2}$$

$$= \frac{2j e^{-j\omega/2}\left(e^{\frac{j\omega}{2}} - e^{-\frac{j\omega}{2}}\right)}{2j}$$

$$= 2j\, e^{-j\omega/2} \sin\left(\frac{\omega}{2}\right)$$

ve $j = e^{-j\frac{3\pi}{2}}$

$$\Rightarrow H(e^{j\omega}) = 2\, e^{-j\left(\omega/2 + \frac{3\pi}{2}\right)} \sin\left(\frac{\omega}{2}\right)$$

$$\left|H(e^{j\omega})\right| = \left|2 \sin\left(\frac{\omega}{2}\right)\right|$$

$$\angle H(e^{j\omega}) = \begin{cases} -\frac{\omega}{2} - \frac{3\pi}{2} & \text{eğer} \quad 0 \le \omega < \frac{\pi}{2} \quad \text{veya} \quad \pi \le \omega < \frac{3\pi}{2} \\ -\frac{\omega}{2} - \frac{3\pi}{2} + \pi & \text{eğer} \quad \frac{\pi}{2} \le \omega < \pi \quad \text{veya} \quad \frac{3\pi}{2} \le \omega < 2\pi \end{cases}$$